From the 2018 Administration

AP® CollegeBoard

AP Calculus AB

Practice Exam

CALCULUS AB

SECTION I, Part A

Time—1 hour

Number of questions—30

NO CALCULATOR IS ALLOWED FOR THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

In this exam:

(1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).

1. If f is the function given by $f(x) = \frac{4}{x} + 5x - 1$, then $f'(2) =$

(A) 4 (B) 6 (C) 7 (D) 11

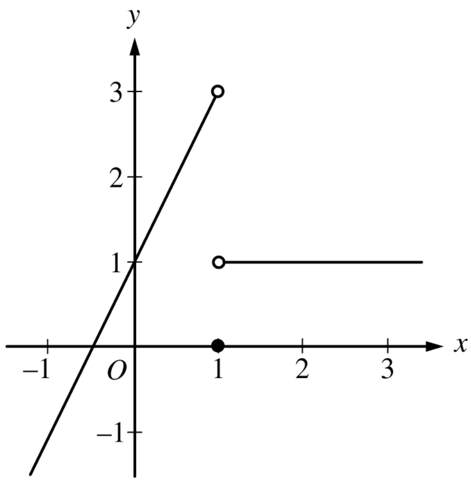
2. $\int 6e^{3x}\,dx =$

(A) $2e^{3x} + C$

(B) $6e^{3x} + C$

(C) $18e^{3x} + C$

(D) $\dfrac{6e^{3x+1}}{3x+1} + C$



Graph of f

3. The graph of $y = f(x)$ is shown above. What is $\lim_{x \to 1} f(x)$?

(A) 0 (B) 1 (C) 3 (D) The limit does not exist.

GO ON TO THE NEXT PAGE.

4. If $f'(x) = 3x^2 + 2x$ and $f(2) = 3$, then $f(1) =$

(A) -10 (B) -7 (C) 10 (D) 13

t (minutes)	0	5	10	15
$R(t)$ (people per minute)	100	100	75	55

5. During an evacuation drill, people leave a building at a rate of $R(t)$ people per minute, where t is the number of minutes since the start of the drill. Selected values of $R(t)$ are shown in the table above. Using a right Riemann sum with three subintervals and data from the table, what is the approximation of the number of people who leave the building during the first 15 minutes of the evacuation drill?

(A) 230 (B) 1150 (C) 1375 (D) 2075

6. If $y = x^2\left(e^x - 1\right)$, then $\dfrac{dy}{dx} =$

(A) $2xe^x$

(B) $2xe^x - 2x$

(C) $x^2e^x + 2xe^x - 2x$

(D) $x^2e^x + 2xe^x - x^2 - 2x$

---

7. A particle moves along the x-axis so that at any time t, $t \geq 0$, its acceleration is $a(t) = -4\sin(2t)$. If the velocity of the particle at $t = 0$ is $v(0) = 7$ and its position at $t = 0$ is $x(0) = 0$, then its position at time t is $x(t) =$

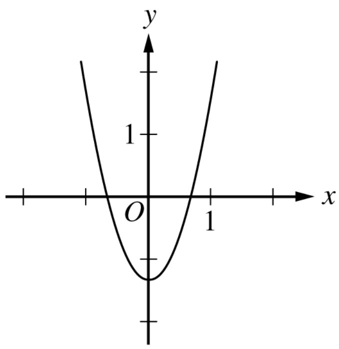
(A) $\sin(2t) + 5t$

(B) $\sin(2t) + 7t$

(C) $\sin(2t) + 9t$

(D) $16\sin(2t) + 7t$

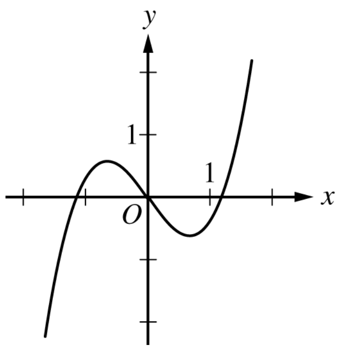
GO ON TO THE NEXT PAGE.



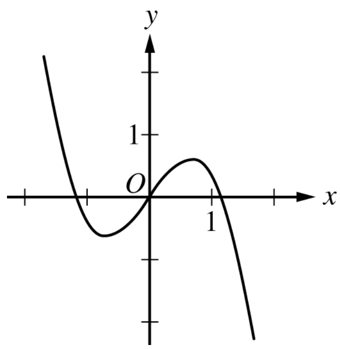
Graph of f''

8. The graph of f'', the second derivative of the function f, is shown above. Which of the following could be the graph of f ?

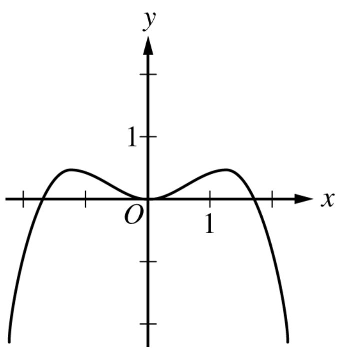
(A)



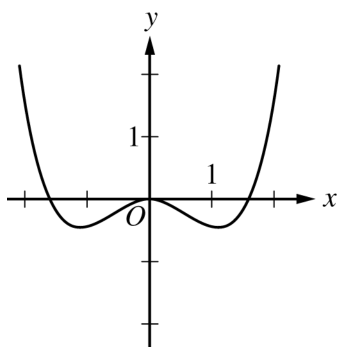
(B)



(C)



(D)



9. When $x = 2e$, $\lim_{h \to 0} \dfrac{\ln(x+h) - \ln(x)}{h}$ is

(A) $\dfrac{1}{2e}$ (B) 1 (C) $\ln(2e)$ (D) nonexistent

---

10. If $\dfrac{dy}{dx} = x^4 - 2x^3 + 3x - 1$, then $\dfrac{d^3y}{dx^3}$ evaluated at $x = 2$ is

(A) 11 (B) 24 (C) 26 (D) 125

$$f(x) = \begin{cases} x^2 & \text{for } x < 0 \\ -1 & \text{for } x = 0 \\ x & \text{for } x > 0 \end{cases}$$

11. Let f be the function defined above. What is $\int_{-1}^{1} f(x)\,dx$?

(A) $\frac{5}{6}$ (B) $\frac{2}{3}$ (C) $-\frac{1}{6}$ (D) nonexistent

GO ON TO THE NEXT PAGE.

12. Given that $3x - \tan y = 4$, what is $\dfrac{dy}{dx}$ in terms of y ?

(A) $\dfrac{dy}{dx} = 3\sin^2 y$

(B) $\dfrac{dy}{dx} = 3\cos^2 y$

(C) $\dfrac{dy}{dx} = 3\cos y \cot y$

(D) $\dfrac{dy}{dx} = \dfrac{3}{1 + 9y^2}$

GO ON TO THE NEXT PAGE.

13. For time $t \geq 1$, the position of a particle moving along the x-axis is given by $p(t) = \sqrt{t} - 2$. At what time t in the interval $1 \leq t \leq 16$ is the instantaneous velocity of the particle equal to the average velocity of the particle over the interval $1 \leq t \leq 16$?

(A) 1 (B) $\frac{121}{25}$ (C) $\frac{25}{4}$ (D) 25

---

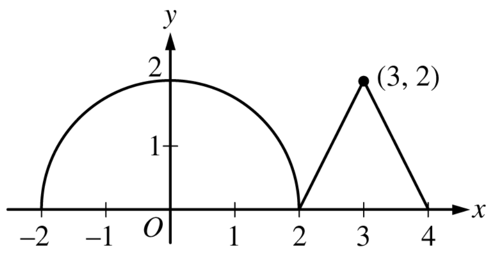
14. If f is a differentiable function and $y = \sin\left(f\left(x^2\right)\right)$, what is $\frac{dy}{dx}$ when $x = 3$?

(A) $\cos(f'(9))$

(B) $6\cos(f(9))$

(C) $f'(9)\cos(f(9))$

(D) $6f'(9)\cos(f(9))$



Graph of g'

15. The graph of g', the first derivative of the function g, consists of a semicircle of radius 2 and two line segments, as shown in the figure above. If $g(0) = 1$, what is $g(3)$?

(A) $\pi + 1$ (B) $\pi + 2$ (C) $2\pi + 1$ (D) $2\pi + 2$

16. Let f be the function given by $f(x) = x^3 - 6x^2 - 15x$. What is the maximum value of f on the interval $[0, 6]$?

(A) 0 (B) 5 (C) 6 (D) 8

17. $\displaystyle\int \frac{1}{x^2 + 4x + 5}\, dx =$

(A) $\arctan(x + 2) + C$

(B) $\arcsin(x + 2) + C$

(C) $\ln\left|x^2 + 4x + 5\right| + C$

(D) $\dfrac{1}{\frac{1}{3}x^3 + 2x^2 + 5x} + C$

---

18. Let f be the function defined by $f(x) = \sqrt[3]{x}$. What is the approximation for $f(10)$ found by using the line tangent to the graph of f at the point $(8, 2)$?

(A) $\dfrac{11}{6}$ (B) $\dfrac{25}{12}$ (C) $\dfrac{13}{6}$ (D) $\dfrac{7}{3}$

19. $\lim_{x \to 0} \dfrac{4x^2}{e^{4x} - 4x - 1}$ is

(A) 0 (B) $\dfrac{1}{2}$ (C) 8 (D) nonexistent

---

20. Let g be a twice-differentiable, increasing function of t. If $g(0) = 20$ and $g(10) = 220$, which of the following must be true on the interval $0 < t < 10$?

(A) $g'(t) = 0$ for some t in the interval.

(B) $g'(t) = 20$ for some t in the interval.

(C) $g''(t) = 0$ for some t in the interval.

(D) $g''(t) > 0$ for all t in the interval.

21. $\dfrac{d}{dx}\displaystyle\int_{e}^{x^3} \ln\left(t^2 + 1\right)\, dt =$

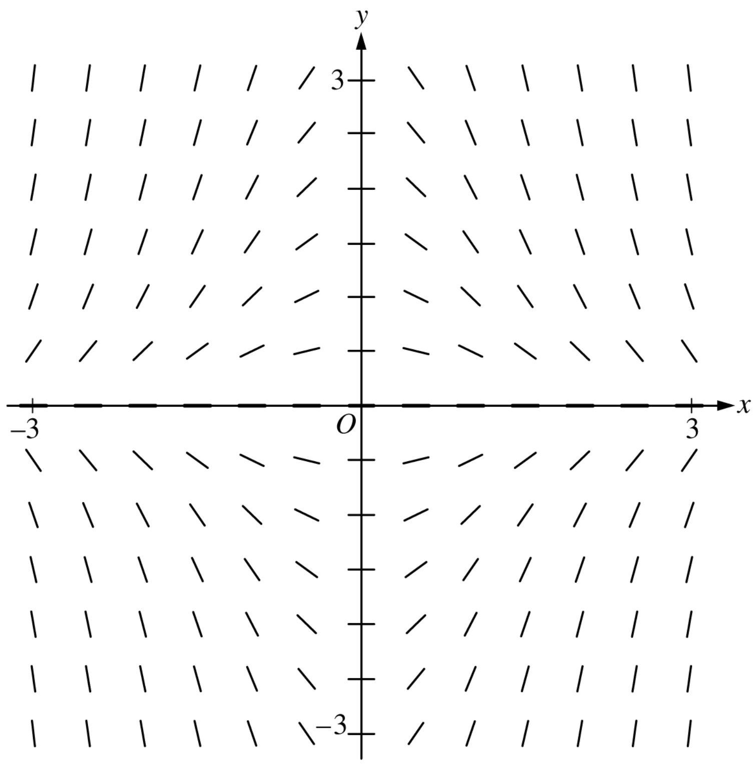
(A) $\ln\left(x^6 + 1\right)$

(B) $3x^2 \ln\left(x^2 + 1\right)$

(C) $3x^2 \ln\left(x^6 + 1\right)$

(D) $\ln\left(x^6 + 1\right) - \ln\left(e^2 + 1\right)$

GO ON TO THE NEXT PAGE.



22. Shown above is a slope field for which of the following differential equations?

(A) $\dfrac{dy}{dx} = \dfrac{x}{y}$

(B) $\dfrac{dy}{dx} = -\dfrac{x}{y}$

(C) $\dfrac{dy}{dx} = xy$

(D) $\dfrac{dy}{dx} = -xy$

GO ON TO THE NEXT PAGE.

23. Using the substitution $u = x + 1$, $\int \frac{x}{\sqrt{x+1}}\,dx$ is equivalent to

(A) $\int \frac{1}{u+1}\,du$ (B) $\int u^{-1/2}\,du$ (C) $\int \left(u^{1/2} - u^{-1/2}\right)du$ (D) $(u-1)\int u^{-1/2}\,du$

---

24. Let f be the function given by $f(x) = \frac{2x^2 + 14x - 16}{x^2 - 9x + 8}$. For what values of x does f have a removable discontinuity?

(A) 1 only (B) 8 only (C) -8 and 1 (D) 1 and 8

25. Which of the following is a solution to the differential equation $y'' - 4y = 0$?

(A) $y = e^{2x}$ (B) $y = 2e^{x}$ (C) $y = \sin(2x)$ (D) $y = \cos(2x)$

GO ON TO THE NEXT PAGE.

x	10	11	12	13	14
$f(x)$	5	2	3	6	5

26. The table above gives values of the continuous function f at selected values of x. If f has exactly two critical points on the open interval $(10, 14)$, which of the following must be true?

(A) $f(x) > 0$ for all x in the open interval $(10, 14)$.

(B) $f'(x)$ exists for all x in the open interval $(10, 14)$.

(C) $f'(x) < 0$ for all x in the open interval $(10, 11)$.

(D) $f'(12) \neq 0$

GO ON TO THE NEXT PAGE.

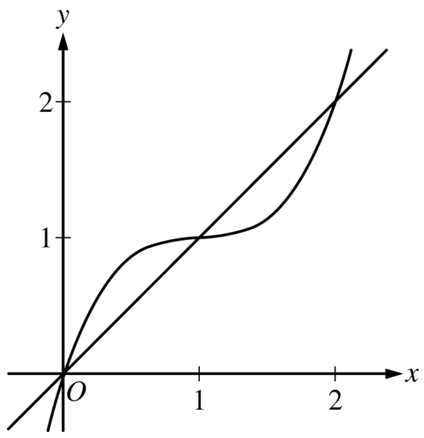
27. The positive variables p and c change with respect to time t. The relationship between p and c is given by the equation $p^2 = (20 - c)^3$. At the instant when $\frac{dp}{dt} = 41$ and $c = 15$, what is the value of $\frac{dc}{dt}$?

(A) $-\frac{82}{75}$ (B) $-\frac{2\sqrt{5}}{3}$ (C) $-\frac{3\sqrt{5}}{2}$ (D) $-\frac{82\sqrt{5}}{15}$

---

28. $\lim_{x \to -\infty} \frac{3 + 2^x}{4 - 5^x}$ is

(A) $-\frac{2}{5}$ (B) 0 (C) $\frac{3}{4}$ (D) nonexistent



29. The graphs of the function g and the line $y = x$ are shown in the figure above. The graphs intersect at the points $(0, 0)$, $(1, 1)$, and $(2, 2)$. Which of the following expressions give the area enclosed by the graphs?

I. $\left| \int_0^2 (x - g(x))\, dx \right|$

II. $\int_0^2 \left| x - g(x) \right| dx$

III. $\int_0^1 (g(x) - x)\, dx + \int_1^2 (x - g(x))\, dx$

(A) II only

(B) III only

(C) I and II only

(D) II and III only

30. A student attempts to solve the differential equation $\frac{dy}{dx} = xy^3$ with the initial condition that $y = 2$ when $x = 0$. The steps of the student's solution are shown below. In which of the following steps does an error first appear?

Step 1: $\int \frac{1}{y^3}\, dy = \int x\, dx$

Step 2: $\ln\left|y^3\right| = \frac{x^2}{2} + C$

Step 3: $\left|y^3\right| = Ke^{x^2/2}$

Step 4: $\left|y^3\right| = 4e^{x^2/2}$

Step 5: $y = \sqrt[3]{4e^{x^2/2}}$

(A) Step 1 (B) Step 2 (C) Step 3 (D) Step 4

END OF PART A

IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON PART A ONLY.

DO NOT GO ON TO PART B UNTIL YOU ARE TOLD TO DO SO.

CALCULUS AB

SECTION I, Part B

Time—45 minutes

Number of questions—15

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–90.

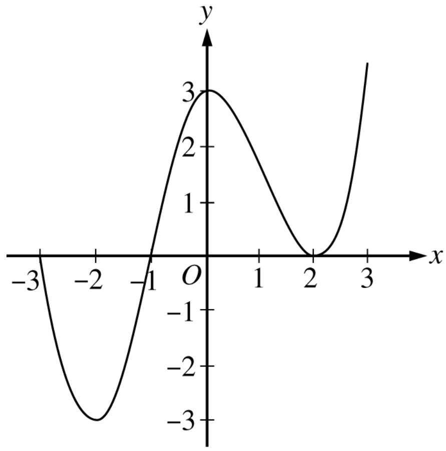
YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

(1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.

(2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).



Graph of f'

76. The graph of f', the derivative of the function f, is shown above for $-3 \le x \le 3$. On what intervals is f increasing?

(A) $[-3, -1]$ only

(B) $[-1, 3]$

(C) $[-2, 0]$ and $[2, 3]$

(D) $[-3, -1]$ and $[1, 3]$

77. The rate at which water leaks from a tank, in gallons per hour, is modeled by R, a differentiable function of the number of hours after the leak is discovered. Which of the following is the best interpretation of $R'(3)$?

(A) The amount of water, in gallons, that has leaked out of the tank during the first three hours after the leak is discovered

(B) The amount of change, in gallons per hour, in the rate at which water is leaking during the three hours after the leak is discovered

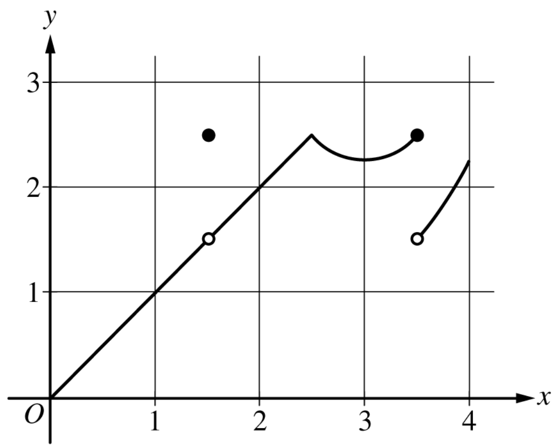
(C) The rate at which water leaks from the tank, in gallons per hour, three hours after the leak is discovered

(D) The rate of change of the rate at which water leaks from the tank, in gallons per hour per hour, three hours after the leak is discovered

---

78. A particle moves along the x-axis. The velocity of the particle at time t is given by $v(t) = \dfrac{4}{t^3 + 1}$. If the position of the particle is $x = 1$ when $t = 2$, what is the position of the particle when $t = 4$?

(A) 0.617 (B) 0.647 (C) 1.353 (D) 5.713



Graph of f

79. The graph of the function f is shown above. Of the following intervals, on which is f continuous but not differentiable?

(A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) $(3, 4)$

GO ON TO THE NEXT PAGE.

80. The first derivative of the function f is defined by $f'(x) = \left(x^2 + 1\right)\sin(3x - 1)$ for $-1.5 < x < 1.5$. On which of the following intervals is the graph of f concave up?

(A) $(-1.5, -1.341)$ and $(-0.240, 0.964)$

(B) $(-1.341, -0.240)$ and $(0.964, 1.5)$

(C) $(-0.714, 0.333)$ and $(1.381, 1.5)$

(D) $(-1.5, -0.714)$ and $(0.333, 1.381)$

---

81. During a rainfall, the depth of water in a rain gauge increases at a rate modeled by $R(t) = 0.5 + t\cos\left(\frac{\pi t^3}{80}\right)$, where t is the time in hours since the start of the rainfall and $R(t)$ is measured in centimeters per hour. How much did the depth of water in the rain gauge increase from $t = 0$ to $t = 3$ hours?

(A) 1.233 cm (B) 1.466 cm (C) 1.966 cm (D) 5.401 cm

82. Let f be a function such that $f(1) = -2$ and $f(5) = 7$. Which of the following conditions ensures that $f(c) = 0$ for some value c in the open interval $(1, 5)$?

(A) $\int_1^5 f(x)\,dx$ exists.

(B) f is increasing on the closed interval $[1, 5]$.

(C) f is continuous on the closed interval $[1, 5]$.

(D) f is defined for all values of x in the closed interval $[1, 5]$.

---

83. The acceleration of a particle moving along the x-axis is given by $a(t) = (t - 8)\sin t$ for $0 \le t \le 8$. At what value of t is the particle's velocity decreasing most rapidly?

(A) 0 (B) 1.420 (C) 3.142 (D) 4.439

84. If the average value of the function f over the closed interval $[2, 4]$ is 3 and if $f(x) \geq 0$ for all x in $[2, 4]$, what is the area of the region enclosed by the graph of $y = f(x)$, the lines $x = 2$ and $x = 4$, and the x-axis?

(A) 12 (B) 6 (C) 3 (D) $\frac{3}{2}$

x	–0.1	–0.01	–0.001	0.001	0.01	0.1
$f(x)$	–0.1054	–0.0101	–0.001	0.001	0.0099	0.0953

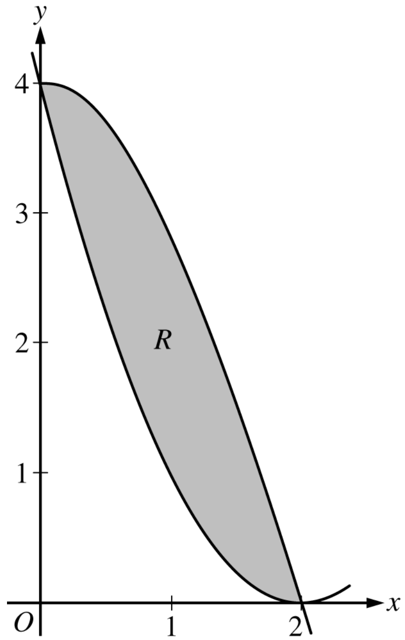
85. The function f is continuous and increasing for $x > -1$. The table above gives values of f at selected values of x. Of the following, which is the best approximation for $\lim_{x \to 0} e^{-2f(x)}$?

(A) -2

(B) 0

(C) 1

(D) The limit does not exist.



86. Let R be the region in the first quadrant bounded by the graphs of $y = 4\cos\left(\frac{\pi x}{4}\right)$ and $y = (x-2)^2$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the x-axis is an isosceles right triangle with a leg in region R. What is the volume of the solid?

(A) 1.775 (B) 3.549 (C) 4.800 (D) 5.575

87. Let f and g be continuous functions. If $\int_2^6 f(x)\,dx = 5$ and $\int_6^2 g(x)\,dx = 7$, then $\int_2^6 (3f(x) + g(x))\,dx =$

(A) -6 (B) 8 (C) 22 (D) 36

---

88. Let f be a twice-differentiable function such that $f''(x) < 0$ for all x. The graph of $y = S(x)$ is the secant line passing through the points $(3, f(3))$ and $(5, f(5))$. The graph of $y = T(x)$ is the line tangent to the graph of f at $x = 4$. Which of the following is true?

(A) $f(4.2) < S(4.2) < T(4.2)$

(B) $f(4.2) < T(4.2) < S(4.2)$

(C) $S(4.2) < f(4.2) < T(4.2)$

(D) $T(4.2) < f(4.2) < S(4.2)$

89. The number of insects in a certain population at time t days is modeled by the function P with first derivative $P'(t) = 0.3t^2 + 12t + 210$. At time $t = 0$, the number of insects in the population is 40. Which of the following statements are true?

I. At time $t = 10$, the number of insects in the population is 2840.

II. At time $t = 10$, the number of insects in the population is increasing at a rate of 360 insects per day.

III. At time $t = 10$, the rate of change of the number of insects in the population is increasing at a rate of 18 insects per day per day.

(A) I only (B) II only (C) III only (D) I, II, and III

GO ON TO THE NEXT PAGE.

x	3	7
$h(x)$	7	22
$h'(x)$	5	10

90. Selected values of the increasing function h and its derivative h' are shown in the table above. If g is a differentiable function such that $h(g(x)) = x$ for all x, what is the value of $g'(7)$?

(A) $-\frac{1}{10}$ (B) $\frac{1}{10}$ (C) $\frac{1}{5}$ (D) $\frac{7}{5}$

GO ON TO THE NEXT PAGE.

CALCULUS AB

SECTION II, Part A

Time—30 minutes

Number of questions—2

A GRAPHING CALCULATOR IS REQUIRED FOR THESE QUESTIONS.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

1. Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g, where $g(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Selected values of $g(t)$ are given in the table above.

(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

Do not write beyond this border.

Continue question 1 on page 5.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Do not write beyond this border.

2. A snail is traveling along a straight path. The snail's velocity can be modeled by $v(t) = 1.4 \ln\left(1 + t^2\right)$ inches per minute for $0 \le t \le 15$ minutes.

(a) Find the acceleration of the snail at time $t = 5$ minutes.

(b) What is the displacement of the snail over the interval $0 \le t \le 15$ minutes?

Do not write beyond this border.

Continue question 2 on page 7.

(c) At what time t, $0 \le t \le 15$, is the snail's instantaneous velocity equal to its average velocity over the interval $0 \le t \le 15$?

(d) An ant arrives at the snail's starting position at time $t = 12$ minutes and follows the snail's path. During the interval $12 \le t \le 15$ minutes, the ant travels in the same direction as the snail with a constant acceleration of 2 inches per minute per minute. The ant catches up to the snail at time $t = 15$ minutes. The ant's velocity at time $t = 12$ is B inches per minute. Find the value of B.

Do not write beyond this border.

GO ON TO THE NEXT PAGE.

CALCULUS AB

SECTION II, Part B

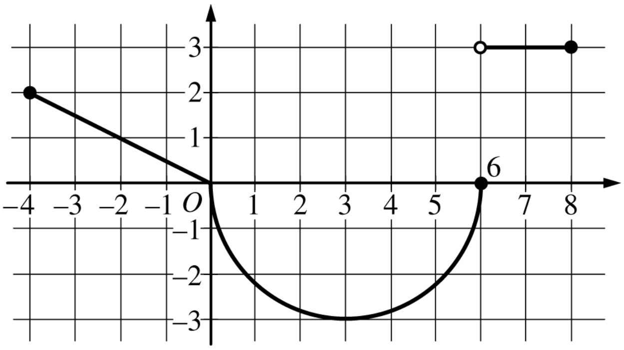
Time—1 hour

Number of questions—4

NO CALCULATOR IS ALLOWED FOR THESE QUESTIONS.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

NO CALCULATOR ALLOWED



Graph of g

3. The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t)\,dt$.

(a) Find $f(7)$ and $f'(7)$.

(b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.

Do not write beyond this border.

Do not write beyond this border.

Continue question 3 on page 15.

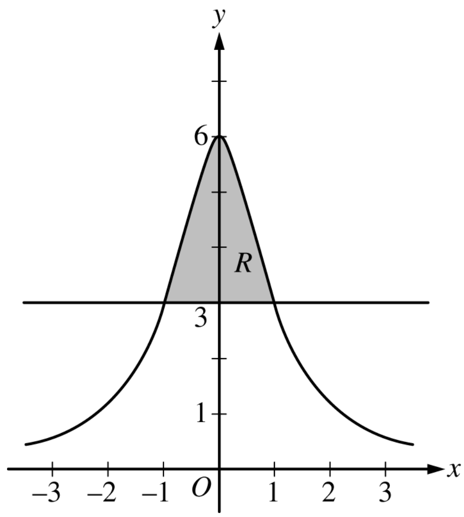
(c) For each of $\lim_{x \to 0^-} g'(x)$ and $\lim_{x \to 0^+} g'(x)$, find the value or state that it does not exist.

(d) Find $\lim_{x \to -2} \dfrac{f(x) + 7}{e^{3x+6} - 1}$.

Do not write beyond this border.

GO ON TO THE NEXT PAGE.

NO CALCULATOR ALLOWED



4. Let f be the function defined by $f(x) = \dfrac{6}{1 + x^2}$. Let R be the shaded region bounded by the graph of f and the horizontal line $y = 3$, as shown in the figure above.

(a) Find the area of R.

Do not write beyond this border.

Continue question 4 on page 17.

(b) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

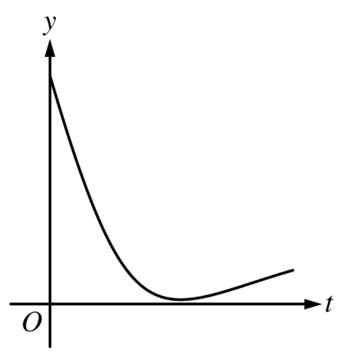
(c) Let $h(x)$ be the vertical distance between the point $(x, f(x))$ and the horizontal line $y = 3$. Find the rate of change of $h(x)$ with respect to x at $x = 2$.

NO CALCULATOR ALLOWED

5. During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction $(t = 0)$, there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\dfrac{dy}{dt} = -0.02y^2$.

(a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.

(b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



Do not write beyond this border.

Continue question 5 on page 19.

(c) Find an expression for $y = f(t)$ by solving the differential equation $\frac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.

(d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

NO CALCULATOR ALLOWED

6. Consider the curve given by the equation $2(x - y) = 3 + \cos y$. For all points on the curve, $\frac{2}{3} \le \frac{dy}{dx} \le 2$.

(a) Show that $\frac{dy}{dx} = \frac{2}{2 - \sin y}$.

(b) For $-\frac{\pi}{2} < y < \frac{\pi}{2}$, there is a point P on the curve through which the line tangent to the curve has slope 1. Find the coordinates of the point P.

Do not write beyond this border.

Do not write beyond this border.

Continue question 6 on page 21.

NO CALCULATOR ALLOWED

(c) Determine the concavity of the curve at points for which $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Give a reason for your answer.

(d) Let $y = f(x)$ be a function, defined implicitly by $2(x - y) = 3 + \cos y$, that is continuous on the closed interval $[2, 2.1]$ and differentiable on the open interval $(2, 2.1)$. Use the Mean Value Theorem on the interval $[2, 2.1]$ to show that $\frac{1}{15} \le f(2.1) - f(2) \le \frac{1}{5}$.

Do not write beyond this border.

Do not write beyond this border.

GO ON TO THE NEXT PAGE.

Answer Key for AP Calculus AB
Practice Exam, Section I

Question 1: A

Question 2: A

Question 3: D

Question 4: B

Question 5: B

Question 6: C

Question 7: A

Question 8: D

Question 9: A

Question 10: B

Question 11: A

Question 12: B

Question 13: C

Question 14: D

Question 15: B

Question 16: A

Question 17: A

Question 18: C

Question 19: B

Question 20: B

Question 21: C

Question 22: D

Question 23: C

Question 24: A

Question 25: A

Question 26: D

Question 27: D

Question 28: C

Question 29: D

Question 30: B

Question 76: B

Question 77: D

Question 78: C

Question 79: C

Question 80: A

Question 81: D

Question 82: C

Question 83: B

Question 84: B

Question 85: C

Question 86: A

Question 87: B

Question 88: C

Question 89: D

Question 90: C

2018 AP Calculus AB
Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
1	2.1C	2.1C1	Implementing algebraic/ computational processes	Building notational fluency	A	76
2	3.3B(a)	3.3B1	Implementing algebraic/ computational processes	Building notational fluency	A	65
3	1.1B	1.1B1	Connecting multiple representations	Reasoning with definitions and theorems	D	84
4	3.3B(b)	3.3B2	Implementing algebraic/ computational processes	Building notational fluency	B	79
5	3.2B	3.2B2	Implementing algebraic/ computational processes	Connecting multiple representations	B	81
6	2.1C	2.1C3	Connecting concepts	Implementing algebraic/ computational processes	C	87
7	3.5A	3.5A1	Connecting concepts	Implementing algebraic/ computational processes	A	44
8	2.2A	2.2A3	Connecting concepts	Connecting multiple representations	D	60
9	2.1A	2.1A3	Reasoning with definitions and theorems	Building notational fluency	A	47
10	2.1D	2.1D1	Implementing algebraic/ computational processes	Building notational fluency	B	90
11	3.2C	3.2C3	Implementing algebraic/ computational processes	Connecting concepts	A	37
12	2.1C	2.1C5	Implementing algebraic/ computational processes	Connecting concepts	B	50
13	2.3C	2.3C1	Implementing algebraic/ computational processes	Connecting concepts	C	68
14	2.1C	2.1C4	Implementing algebraic/ computational processes	Connecting concepts	D	63
15	3.2C	3.2C1	Connecting multiple representations	Implementing algebraic/ computational processes	B	45
16	2.2A	2.2A1	Implementing algebraic/ computational processes	Building notational fluency	A	43
17	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Building notational fluency	A	33
18	2.3B	2.3B2	Connecting concepts	Implementing algebraic/ computational processes	C	56
19	1.1C	1.1C3	Implementing algebraic/ computational processes	Reasoning with definitions and theorems	B	46
20	2.4A	2.4A1	Reasoning with definitions and theorems	Connecting concepts	B	54
21	3.3A	3.3A2	Implementing algebraic/ computational processes	Building notational fluency	C	49
22	2.3F	2.3F1	Connecting multiple representations	Connecting concepts	D	71
23	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Connecting concepts	C	25
24	1.2A	1.2A3	Implementing algebraic/ computational processes	Reasoning with definitions and theorems	A	56
25	2.3E	2.3E2	Implementing algebraic/ computational processes	Building notational fluency	A	51
26	2.2A	2.2A1	Connecting multiple representations	Reasoning with definitions and theorems	D	22
27	2.3C	2.3C2	Implementing algebraic/ computational processes	Connecting concepts	D	45
28	1.1D	1.1D1	Building notational fluency	Implementing algebraic/computational processes	C	26
29	3.4D	3.4D1	Connecting multiple representations	Connecting concepts	D	34
30	3.5A	3.5A2	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	B	46

2018 AP Calculus AB
Question Descriptors and Performance Data

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
76	2.2A	2.2A3	Connecting multiple representations	Connecting concepts	B	81
77	2.3A	2.3A2	Connecting concepts	Building notational fluency	D	69
78	3.4C	3.4C1	Implementing algebraic/ computational processes	Connecting concepts	C	66
79	2.2B	2.2B1	Connecting multiple representations	Connecting concepts	C	80
80	2.2A	2.2A1	Connecting concepts	Implementing algebraic/ computational processes	A	57
81	3.4E	3.4E1	Implementing algebraic/ computational processes	Connecting concepts	D	78
82	1.2B	1.2B1	Reasoning with definitions and theorems	Building notational fluency	C	68
83	2.3C	2.3C1	Connecting concepts	Implementing algebraic/ computational processes	B	56
84	3.4B	3.4B1	Connecting concepts	Implementing algebraic/ computational processes	B	71
85	1.1B	1.1B1	Connecting multiple representations	Implementing algebraic/ computational processes	C	60
86	3.4D	3.4D2	Implementing algebraic/ computational processes	Connecting multiple representations	A	43
87	3.2C	3.2C2	Reasoning with definitions and theorems	Connecting concepts	B	79
88	2.3B	2.3B1	Connecting concepts	Reasoning with definitions and theorems	C	41
89	2.3D	2.3D1	Implementing algebraic/ computational processes	Connecting concepts	D	68
90	2.1C	2.1C6	Connecting concepts	Connecting multiple representations	C	36

2018 AP Calculus AB
Question Descriptors and Performance Data

Free-Response Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus	Mean
1	2.1B|2.3A|2.3A|3.2B|3.3B(b)|3.4A	2.1B1|2.3A1|2.3A2|3.2B2|3.3B2|3.4A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	4.98
2	2.3C|3.3B(b)|3.4B|3.4C	2.3C1|3.3B2|3.4B1|3.4C1	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Building notational fluency|Communicating	3.31
3	1.1A(b)|1.1B|1.1C|2.2A|3.2C|3.2C|3.3A	1.1A3|1.1B1|1.1C3|2.2A1|3.2C1|3.2C3|3.3A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	2.34
4	2.1C|2.3A|3.3B(b)|3.4D|3.4D	2.1C4|2.3A2|3.3B2|3.4D1|3.4D2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	2.63
5	2.1C|2.1D|2.2A|2.3B|3.5A	2.1C4|2.1D1|2.2A1|2.3B2|3.5A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	2.34
6	1.2B|2.1C|2.1D|2.2A|2.3B|2.4A	1.2B1|2.1C5|2.1D1|2.2A1|2.3B1|2.4A1	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Building notational fluency|Communicating	2.78

AP® CALCULUS AB
2018 SCORING GUIDELINES

Question 1

(a) $g'(3) \approx \dfrac{g(5) - g(1)}{5 - 1} = \dfrac{20.5 - 15.1}{4} = 1.35$ At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.	$2 : \begin{cases} 1 : \text{approximation} \\ 1 : \text{interpretation with units} \end{cases}$
(b) The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet. $\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$ $= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$	$3 : \begin{cases} 1 : \text{integral expression} \\ 1 : \text{right Riemann sum} \\ 1 : \text{approximation} \end{cases}$
(c) $\int_0^8 w(t)\,dt = 99.051497$ The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\,dt = 61.749$ (or 61.748) cubic feet.	$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$
(d) $g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$ Because $g(6) - w(6) > 0,$ the amount of unspoiled grain is increasing at time $t = 6$.	$2 : \begin{cases} 1 : \text{considers } g(6) - w(6) \\ 1 : \text{answer} \end{cases}$

Question 2

(a) $v'(5) = 0.538462$

The acceleration of the snail at time $t = 5$ minutes is 0.538 inches per minute per minute.

1 : answer

(b) $\int_0^{15} v(t)\, dt = 76.043074$

The displacement of the snail over the interval $0 \le t \le 15$ minutes is 76.043 inches.

$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(c) $\frac{1}{15}\int_0^{15} v(t)\, dt = 5.069538$

$1.4\ln\left(1 + t^2\right) = 5.069538 \Rightarrow t = 6.031$ minutes

$2 : \begin{cases} 1 : \text{average value expression} \\ 1 : \text{answer} \end{cases}$

(d) The velocity of the ant at time t, $12 \le t \le 15$, is $\int 2\, dt = 2t + c$ inches per minute for some constant c.

For $12 \le t \le 15$, the displacement of the ant is
$\int_{12}^{15} (2t + c)\, dt = \left(t^2 + ct\right)\Big|_{t=12}^{t=15} = 81 + 3c$ inches.

Thus, $81 + 3c = 76.043074 \Rightarrow c = -1.652309$.

The velocity of the ant at time $t = 12$ is
$B = 2 \cdot 12 - 1.652309 = 22.348$ (or 22.347) inches per minute.

— OR —

The velocity of the ant at time t, $12 \le t \le 15$, is $2(t - 12) + B$ inches per minute.

For $12 \le t \le 15$, the displacement of the ant is
$\int_{12}^{15} (2(t - 12) + B)\, dt = \left((t - 12)^2 + Bt\right)\Big|_{t=12}^{t=15} = 9 + 3B$ inches.

$9 + 3B = 76.043074 \Rightarrow B = 22.348$ (or 22.347) inches per minute

$4 : \begin{cases} 1 : \text{ant's velocity} \\ 1 : \text{ant's displacement} \\ 1 : \text{equation} \\ 1 : \text{answer} \end{cases}$

Question 3

(a) $f(7) = 3 \cdot 7 + \int_0^7 g(t)\,dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$ $f'(7) = 3 + g(7) = 3 + 3 = 6$	$2 : \begin{cases} 1 : f(7) \\ 1 : f'(7) \end{cases}$
(b) On the interval $-4 \le x \le 3$, $f'(x) = 3 + g(x)$. Because $f'(x) \ge 0$ for $-4 \le x \le 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.	2 : answer with justification
(c) $\lim\limits_{x \to 0^-} g'(x) = -\frac{1}{2}$ $\lim\limits_{x \to 0^+} g'(x)$ does not exist.	$2 : \begin{cases} 1 : \text{left-hand limit} \\ 1 : \text{right-hand limit} \end{cases}$
(d) $\lim\limits_{x \to -2} (f(x) + 7) = -6 + \int_0^{-2} g(t)\,dt + 7 = 0$ $\lim\limits_{x \to -2} \left(e^{3x+6} - 1\right) = 0$ Using L'Hospital's Rule, $\lim\limits_{x \to -2} \frac{f(x) + 7}{e^{3x+6} - 1} = \lim\limits_{x \to -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3 + g(-2)}{3} = \frac{3 + 1}{3} = \frac{4}{3}.$	$3 : \begin{cases} 1 : \text{limits equal 0} \\ 1 : \text{applies L'Hospital's Rule} \\ 1 : \text{answer} \end{cases}$ Note: max $1/3$ [1-0-0] if no limit notation attached to a ratio of derivatives

Question 4

(a) $f(x) = 3 \Rightarrow x = -1$ and $x = 1$

$$\int_{-1}^{1} (f(x) - 3)\, dx = \int_{-1}^{1} \left(\frac{6}{1 + x^2} - 3 \right) dx$$
$$= \left(6\tan^{-1} x - 3x \right) \Big|_{-1}^{1}$$
$$= \left(6\tan^{-1} 1 - 3 \right) - \left(6\tan^{-1} (-1) + 3 \right)$$
$$= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right)$$
$$= 3\pi - 6$$

The area of R is $3\pi - 6$.

$$3 : \begin{cases} 1 : \text{integral} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$$

(b) Volume $= \pi \int_{-1}^{1} \left((7 - 3)^2 - (7 - f(x))^2 \right) dx$

$$3 : \begin{cases} 2 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$$

(c) $h(x) = 3 - f(x) = 3 - \dfrac{6}{1 + x^2}$ for $x > 1$

$h'(x) = \dfrac{12x}{\left(1 + x^2\right)^2}$ for $x > 1$

$h'(2) = \dfrac{12 \cdot 2}{5^2} = \dfrac{24}{25}$

$$3 : \begin{cases} 1 : h(x) \\ 1 : h'(x) \\ 1 : h'(2) \end{cases}$$

Question 5

(a) $y'(0) = -0.02\left(10^2\right) = -2$ An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$. $y(2) \approx 10 - 2(2) = 6$ grams	$2 : \begin{cases} 1 : y'(0) \\ 1 : \text{approximation} \end{cases}$
(b) $\dfrac{dy}{dt} = -0.02y^2 \le 0,$ so the graph of f is nonincreasing. The graph of f cannot resemble the given graph because the given graph is increasing on a portion of its domain.	1 : answer with reason
(c) $\displaystyle\int\left(-\frac{1}{y^2}\right) dy = \int 0.02\ dt$ $\dfrac{1}{y} = 0.02t + C$ $\dfrac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$ $\dfrac{1}{y} = 0.02t + 0.1 \Rightarrow y = \dfrac{1}{0.02t + 0.1} = \dfrac{50}{t + 5}$ Note: this solution is valid for $t > -5$.	$4 : \begin{cases} 1 : \text{separation of variables} \\ 1 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ \quad \text{and uses initial condition} \\ 1 : \text{answer} \end{cases}$ Note: max $2/4$ [1-1-0-0] if no constant of integration Note: $0/4$ if no separation of variables
(d) $\dfrac{d^2y}{dt^2} = -0.04y\dfrac{dy}{dt} = -0.04y\left(-0.02y^2\right) = 0.0008y^3$ Because $y > 0$, $0.0008y^3 > 0$. The amount of substance is changing at an increasing rate. — OR — From part (c), $f(t) = \dfrac{50}{t + 5}$, and from context, $t \ge 0$. $f'(t) = \dfrac{-50}{(t + 5)^2}$ and $f''(t) = \dfrac{100}{(t + 5)^3} > 0$ for $t \ge 0$. The amount of substance is changing at an increasing rate.	$2 : \begin{cases} 1 : \dfrac{d^2y}{dt^2} \\ 1 : \text{answer with reason} \end{cases}$

Question 6

(a) $\dfrac{d}{dx}(2(x-y)) = \dfrac{d}{dx}(3+\cos y)$

$2 - 2\dfrac{dy}{dx} = (-\sin y)\dfrac{dy}{dx}$

$2 = (2-\sin y)\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = \dfrac{2}{2-\sin y}$

$2: \begin{cases} 1: \text{implicit differentiation} \\ 1: \text{verification} \end{cases}$

(b) $\dfrac{dy}{dx} = \dfrac{2}{2-\sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$

$2(x-0) = 3+\cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$

Point P has coordinates $(2, 0)$.

$2: \begin{cases} 1: \dfrac{dy}{dx} = 1 \\ 1: \text{answer} \end{cases}$

(c) $\dfrac{d^2y}{dx^2} = \dfrac{-2}{(2-\sin y)^2}(-\cos y)\dfrac{dy}{dx} = \dfrac{4\cos y}{(2-\sin y)^3}$

$\dfrac{d^2y}{dx^2} > 0$ for all y in the interval $-\dfrac{\pi}{2} < y < \dfrac{\pi}{2}$.

Therefore, the curve is concave up for $-\dfrac{\pi}{2} < y < \dfrac{\pi}{2}$.

$3: \begin{cases} 2: \dfrac{d^2y}{dx^2} \\ 1: \text{answer with reason} \end{cases}$

(d) By the Mean Value Theorem, for some value c in the interval $(2, 2.1)$, $f'(c) = \dfrac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\dfrac{2}{3} \le f'(x) \le 2$.

Thus, $\dfrac{2}{3} \le \dfrac{f(2.1)-f(2)}{0.1} \le 2 \Rightarrow \dfrac{1}{15} \le f(2.1)-f(2) \le \dfrac{1}{5}$.

$2: \begin{cases} 1: \text{applies Mean Value Theorem} \\ 1: \text{verification} \end{cases}$